TECHNOLOGY PLAN · FOR STAKEHOLDERS

Dubai Real Estate CRM: what it runs on and what it costs

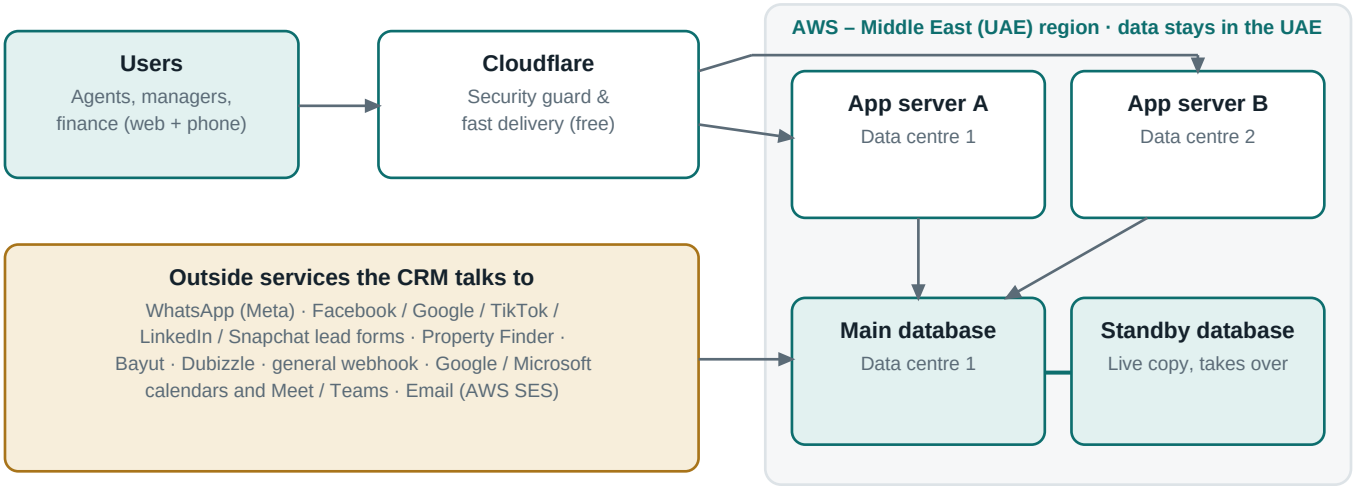
This document explains, in plain language, every technology service the CRM will use, where it runs, why we need it and what it costs each month. The plan follows the developer requirements brief of 29 September 2026: a desktop- and mobile-friendly CRM for Dubai brokerages, sold to companies (B2B) and individual agents (B2C).

PLATFORM COST	BUDGET	BREAK-EVEN	DATA LOCATION
AED 554	**AED 1,000**	**6 users**	**UAE**
per month incl. VAT	AED 446 to spare	at AED 99 per user	AWS UAE region

The plan in one paragraph

The CRM runs on Amazon Web Services (AWS) in its UAE region, so client and lead data never leaves the country. Two app servers in two separate data centres share the work; if one fails, the other carries on. The database keeps a live standby copy that takes over automatically within one to two minutes, and is backed up every day. Free services handle security, login, error alerts and code delivery. Everything is built on standard open-source software, so the system can grow to thousands of users, or move to Azure or our own servers later, without being rebuilt.

How it fits together



Both app servers use the main database. The dashed line shows every change being copied to the standby instantly.

1. Paid services

Prices are list prices in October 2026, converted at 1 USD = 3.6725 AED. Servers and database are on 1-year commitments, which cut their price by roughly a third.

SERVICE	TECHNOLOGY	WHAT IT DOES AND WHY WE NEED IT	WHERE	AED / MONTH
App servers	AWS EC2 (2× t4g.small)	The computers that run the CRM software. Two of them, in two buildings, so one can fail without anyone noticing.	AWS UAE	**73**
Database	AWS RDS PostgreSQL (db.t4g.medium, Multi-AZ)	Where every lead, call, note, invoice and attendance record is kept. A live standby copy takes over automatically if the main one fails.	AWS UAE	**272**
Database storage & backups	AWS gp3 storage + automatic backups	Disk space for the database plus daily backups kept 14 days, so we can restore any moment in that window.	AWS UAE	**18**
Internet addresses	AWS public IPv4 (2)	Lets the servers call WhatsApp, Meta, Google and the portals.	AWS UAE	**26**
File storage	AWS S3	Holds brochures, floor plans, invoice PDFs and attendance photos.	AWS UAE	**7**
Monitoring	AWS CloudWatch	Watches server health and raises an alarm if something looks wrong.	AWS UAE	**11**
Test environment	AWS EC2 Spot, office hours only	A separate copy for testing changes before staff see them (required by the brief).	AWS UAE	**11**
Email sending	AWS SES	Sends reminders, meeting confirmations and reports; receives portal lead emails.	AWS	**11**
Error tracking	Sentry Team	Tells developers instantly when something breaks, with details to fix it fast.	Sentry (EU/US)	**95**
Domain name	.com or .ae	The web address of the CRM.	Registrar	**4**
Card payments	Stripe	Takes subscription payments by card and sends payment updates to the CRM. No monthly fee; about 3% per payment, taken from revenue (section 12).	Stripe	**0**
Total before VAT				**528**
UAE VAT 5%				**26**
Total per month				**AED 554**

2. Free services (AED 0)

These cost nothing at our size. Each has a paid tier we can switch on later if needed.

SERVICE	TECHNOLOGY	WHAT IT DOES AND WHY WE NEED IT
Security & speed	Cloudflare Free + Cloudflare Tunnel	Stands in front of the CRM: blocks attacks, provides the padlock (SSL), serves images fast and sends visitors to whichever server is healthy.
Login & two-factor	AWS Cognito	Secure sign-in with a code from an authenticator app. Free up to 10,000 active users a month.
Background jobs	pg-boss (inside the database)	Sends reminders, escalates untouched leads and retries failed syncs in the background, without an extra server.
Uptime alerts	Better Stack Free	Checks every few minutes that the CRM is reachable and alerts the team if not.
Code storage & delivery	GitHub Free + GitHub Actions	Keeps the source code (owned by the company) and releases tested updates automatically.
Maps	Google Maps Platform (free allowance)	Shows project and unit locations.
Phone notifications	Firebase Cloud Messaging	Pushes reminders and alerts to agents' phones.
General webhook	Built into the CRM	One secure link that any system can send leads to: website, Google Sheets, Excel (Power Automate) or Make.com.
Online meetings	Google Meet / Microsoft Teams	Meeting links created from the agent's connected Google or Microsoft account. Replaces Zoom.
Client booking page	Built into the CRM	Clients pick a free slot in the agent's calendar. Replaces Calendly.
PDFs & Excel	Gotenberg, ExcelJS (open source)	Creates invoice and purchase-order PDFs and Excel report exports on our own servers.
Lead sources	Meta, Google Ads, LinkedIn, TikTok, Snapchat APIs	Bring ad leads straight into the CRM, with campaign and form details. Free to connect; ad spend is separate.
Property portals	Property Finder, Bayut, Dubizzle	Leads arrive through each account's authorized method (partner API, webhook or lead email). Confirm API access per account.
Calendars	Google Calendar, Microsoft 365	Two-way calendar sync for agents. Uses the brokerage's existing accounts.

3. Paid by each brokerage on its own accounts

The brief asks to *connect* these, meaning the brokerage's existing subscriptions. They are not part of our monthly cost.

SERVICE	WHY	TYPICAL COST TO THE BROKERAGE
Zoom, Calendly, respond.io	Not needed. Meetings use free Google Meet / Teams links, clients book through the CRM's own booking page, and WhatsApp connects directly to Meta.	AED 0

4. Software the CRM is built with

All open source: no licence fees, no lock-in to one vendor, and the company owns the code.

PART	TECHNOLOGY	WHY THIS CHOICE
Screens (web & phone)	Next.js (React, TypeScript)	One app for desktop and mobile. Installs on phones like an app, including camera check-in for attendance, without app-store fees.
Business logic	NestJS (Node.js, TypeScript)	Organised into modules: leads, tasks, WhatsApp/email, projects, invoices, attendance, reports, integrations. Any module can become its own service later.
Data	PostgreSQL 16	Reliable, industry-standard database. Keeps each brokerage's data separate, stores original ad-form answers, and powers search and duplicate detection.
Packaging & setup	Docker, Terraform	The whole system is described as code, so test and live environments are identical and can be rebuilt or moved quickly.

5. What stakeholders can count on

Always on	Two servers in two data centres plus a standby database. A failure in one building does not stop the CRM.
Data safety	Daily backups, restore to any moment in the last 14 days, all data encrypted, every change recorded in an audit log.
Security	Two-factor login, role-based permissions from Director to Agent, attack protection by Cloudflare, data kept in the UAE.
Speed	Lead lists open in under a second; heavy reports and Excel exports run in the background so screens stay fast.
Ownership	The company owns the data, source code, database design and documentation, as the brief requires.

6. How cost grows with the business

We only upgrade when a trigger is reached. Each step is a settings change, not a rebuild. Platform cost only; integrations are billed to brokerages (section 10). Growth uses AWS's managed database for simplicity; the lean design used at 1 million users is cheaper but needs more hands-on operation.

STAGE	SIZE	WHAT WE ADD	AED / MONTH
Launch	1 brokerage, up to ~150 users	This plan	≈ **554**
Growth	10 brokerages, ~500 users	Larger servers and database, small cache server	≈ **1,508**
1 million users	Mostly buyers and tenants, ~20,000 active a day	Lean design (section 14): Cloudflare caching, self-run database, self-hosted login and maps, 3-year commitments	≈ **1,281**

7. Why AWS, and not Azure or our own servers

OPTION	MONTHLY COST	VERDICT
AWS, UAE region (recommended)	≈ AED 554	Meets budget with automatic database failover. Widest choice of services as we grow.
Microsoft Azure, UAE North	≈ AED 1,100–1,150	About 3× more because its cheapest database tier has no standby copy. Worth revisiting at large scale: Azure has two UAE regions (Dubai and Abu Dhabi) for in-country disaster recovery and strong ties with government and Microsoft-based enterprises.
Own servers (on-premise)	≈ AED 5,000–12,000	Needs 3 servers, two internet lines, backup power and a system administrator. WhatsApp and ad lead feeds still come from the cloud. Possible later because the software is open source.

8. One-off costs (not monthly)

• Security penetration test before go-live: about AED 15,000–40,000, usually once a year.
• UAE SMS sender ID registration, only if SMS is added later.
• Native iPhone / Android apps, only if needed: Apple AED 364 a year, Google AED 92 once.
• Data migration from the current CRM and staff training (part of the development quote).

9. Assumptions and notes

• Launch size: one brokerage, about 50 users and 5,000 new leads a month, of which about 60% never respond.
• AWS amounts are estimates from public list prices for the UAE region; confirm in the AWS Pricing Calculator before signing.
• WhatsApp rates are indicative UAE rates (about USD 0.015 per reminder, USD 0.0385 per marketing message); confirm on Meta's live rate card.
• Not included: developer salaries, advertising spend, and card fees when collecting subscriptions (Stripe UAE 2.9% + AED 1 per payment).
• Servers and database are on 1-year commitments. Monthly pay-as-you-go would add about AED 150 a month.

10. Third-party integrations and lead rules

WhatsApp is now paid by each customer company directly to Meta: every brokerage or agent connects its own WhatsApp Business account and Meta bills them. Zoom, Calendly and respond.io have been removed. Stripe card fees are taken from revenue (section 12). So our only running cost is the platform. The lead rules below (10-minute rule, re-enquiry, sharing, scoring) run on the servers we already pay for.

ITEM (LAUNCH: 50 USERS, 5,000 LEADS A MONTH)	AED / MONTH
Platform (page 2): servers, database, backups, email, monitoring, error tracking	**528**
WhatsApp messages (paid by each company directly to Meta)	**0**
Card payment fees (Stripe, about 3%, deducted from revenue)	**0**
Total before VAT	**528**
UAE VAT 5%	**26**
Total per month, all-in	**AED 554**

Example of what a brokerage pays Meta itself: a welcome WhatsApp to 5,000 new leads a month is about AED 290, follow-up templates about AED 150. Replies within 24 hours of a customer's message are free.

Lead rules agreed with stakeholders

10-minute rule	A new lead not acted on within 10 minutes (adjustable) moves to the next available agent in the team. If every agent in the team misses it, the lead goes back to the **Admin**, who sees the full history and decides.
Re-enquiry	A repeat enquiry from an existing contact is added to the existing lead and stays with the same agent, who is notified "Re-enquired". No duplicate leads.
Lead sharing	The owner or a manager can share a lead with other users as "View only" or "Can edit".
Focus on potential	Each lead gets a score from 0 to 100 (budget, project match, source quality, engagement). Best leads show first. Leads are never closed automatically; the brokerage team decides.

Meta decides whether a follow-up is billed as a utility or a marketing message; follow-ups are priced here at the higher marketing rate.

11. Development cost and timeline

The CRM is designed and built in-house by the solution architect, so there is **no development cost**. The company still owns the code, database design and documentation, as the brief requires.

PHASE	WHAT IS DELIVERED	WHEN (ONE DEVELOPER)	COST
Phase 1	Core: leads, lead tabs, re-enquiry, sharing, 10-minute rule, tasks & calendar, team hierarchy & permissions, dashboard, reports & Excel export, Excel/CSV import, Meta lead ads, subscriptions, billing, sign-up and owner console	Months 1–5	**AED 0**
Phase 2	Communications & integrations: WhatsApp, email & campaigns, consent/opt-out, Google/LinkedIn/TikTok/Snapchat, Property Finder, Bayut & Dubizzle, general webhook, booking page, calendars, connection screen, Meta reconciliation	Months 6–9	**AED 0**
Phase 3	Projects & units, deals, invoices, purchase orders & payments, commissions, attendance with photo check-in	Months 10–12	**AED 0**
Total	About 12 months; version 1 launches with all three phases (SRS v1.2)		**AED 0**

Keeping a one-person build safe

• Code lives in the company's GitHub account, not on a personal laptop.
• Automated tests and Terraform mean another developer can take over or join quickly.
• Short written documentation for each module, plus the API documentation the brief requires.
• Still budget for an external penetration test before go-live (AED 15,000–40,000, one-off), since nobody should test their own security alone.

Running costs (page 2) start at go-live, about AED 554 a month.

12. Revenue and break-even (example)

Agreed pricing: **AED 99 per user per month** on every plan (Solo, Team; Enterprise at a custom price), excluding 5% VAT. VAT collected is paid to the tax authority, so it is not revenue. Each company pays Meta for its own WhatsApp. Revenue below is after about 3% Stripe card fees.

PAYING USERS	REVENUE	PLATFORM COST	MONTHLY PROFIT
50	**4,802**	**554**	**4,248**
200	**19,206**	**1,100**	**18,106**
500	**48,015**	**1,508**	**46,507**
1,000	**96,030**	**2,714**	**93,316**

All amounts in AED per month, excluding VAT on revenue. Platform cost for 200 and 1,000 users is estimated between the launch and 500-user figures. With no development cost there is no build investment to repay. Profit is before sales, support and marketing staff.

What this means

• Running costs are covered by the first **6 paying users**.
• At **500 users** (about 10 brokerages) the platform earns about AED 46,507 a month after running costs.
• Each extra user adds about AED 90 a month of profit, because running cost per user keeps falling as the platform grows.
• The 14-day trial needs a card, so trials turn into paying users automatically unless cancelled.

13. To confirm before final sign-off

These items turn the estimates in this document into confirmed numbers.

ITEM	ACTION	OWNER	STATUS
AWS running cost	Enter this plan in the AWS Pricing Calculator for the UAE region	Architect	To do
Stripe account	Open a Stripe UAE account under the trade licence; confirm card fee rates	Management	To do
Yearly price	Confirm yearly price (draft AED 990 per user, 2 months free)	Management	To do
Property Finder, Bayut & Dubizzle	Confirm API, webhook or lead-email access for the brokerage's accounts	Brokerage + architect	To do
Ad platform access	Start Meta app review and LinkedIn lead-sync partner approval (takes weeks)	Architect	To do
Open SRS items	Confirm items marked "to confirm" in SRS v1.2	Stakeholder	To do
Terms and privacy	Terms of Service and Privacy Policy for customers (UAE lawyer)	Legal	To do
Data protection	Legal review against the UAE data protection law (PDPL)	Legal	To do
Launch date	Agree the go-live date for version 1 (all phases)	Management + architect	To do
Penetration test	Get a quote from a UAE security firm	Architect	To do

14. Design for 1 million users at low cost

A million users in Dubai real estate would be mostly buyers and tenants browsing listings, with a few thousand agents doing the heavy work. If about 2% are active each day (20,000 people), the servers receive only around 12 requests a second, because Cloudflare answers most page views from its cache. The design below keeps the platform close to the AED 1,000 budget while data stays in the UAE.

DESIGN CHOICE	WHY IT SAVES	WHERE	AED / MONTH
Cloudflare caching for all public pages and images	About 95% of visits never reach our servers	Cloudflare	**0**
Property images and brochures in Cloudflare R2 (2 TB)	No fee each time an image is viewed. Private files (invoices, attendance photos) stay in AWS UAE	Cloudflare	**110**
2 app servers, 3-year commitment	About 60% cheaper than monthly pricing; still two data centres	AWS UAE	**260**
PostgreSQL run by us on 2 servers with a live standby (Patroni)	About half the cost of AWS's managed database; automatic failover kept	AWS UAE	**735**
Self-hosted login (Keycloak)	Cognito would charge about AED 54,000 a month at 1 million users	AWS UAE	**0**
OpenStreetMap maps (MapLibre)	No fee per map view; Google Maps would charge per load	AWS UAE	**0**
Phone push notifications first, email only when needed	Push is free; email is charged per message	Firebase / AWS	**75**
Backups to cold storage; old activity archived	Keeps the live database small and cheap	AWS UAE	**40**
Total before VAT			**1,220**
UAE VAT 5%			**61**
Platform cost per month at 1 million users			**AED 1,281**

What design cannot remove

• **WhatsApp:** Meta charges for every message. One message to each of a million people is about AED 55,000, so messaging is billed to the brokerage that sends it.
• **Heavy agent use:** if a million people all worked in the CRM like agents, data, email and server load would grow far beyond this budget.
• **Hosting outside the UAE** (for example dedicated servers in Germany) could bring the platform under AED 1,000, but client data would leave the UAE.

Trade-off: we run the database ourselves, including failover and backups. This is reliable with good automation and monitoring, but adds operating work for a one-person team.

15. Coverage of the requirements brief

Every section of the developer requirements brief (29 September 2026), the technology that delivers it, and the phase it ships in.

BRIEF SECTION	HOW IT IS DELIVERED	PHASE
1 Layout & navigation	Next.js screens: left menu, lead tabs, global search, saved filters, configurable columns, bulk actions, role-based export	1
2 Leads	Leads module; PostgreSQL stores original form answers and source IDs; duplicate detection and merge; full timeline; configurable stages with reasons	1
3 Tasks, calls, meetings	Tasks module; reminders; 10-minute rule (next agent, then Admin); Google Meet / Teams links; built-in booking page; Google / Microsoft calendar sync; client time zones	1 (tasks) · 2
4 Team & permissions	Director → Agent hierarchy, Admin and Finance roles; per-action permissions; round-robin and rule-based assignment; reassignment log	1
5 WhatsApp & email	WhatsApp Cloud API directly with Meta, AWS SES; templates, bulk audiences, delivery statuses; consent and opt-out suppression	2
6 Projects & listings	Developer, project, building and unit records; images and brochures in file storage; unit status and price history	3
7 Invoices & purchase orders	Finance module; PDF generation (Gotenberg); approval workflow, payments, balances, commissions; full audit	3
8 Reports & exports	Reports module; Excel/CSV exports matching on-screen totals; Meta lead reconciliation; export log	1 · 2
9 Attendance	Phone check-in with live photo and optional location; office hours and leave rules; correction approvals with audit trail	3
10 Dashboard	Role-based dashboards, pie, funnel, column and line charts, KPI cards; every number opens its records	1
11 Configuration & integrations	Connection screen per source: status, last sync, failures, test, retry; general webhook with field mapping; Dubizzle added; portal access confirmed per account	2
12 Security & delivery	Two-factor login, encryption, daily backups, audit logs, separate staging; company owns code, schema, API docs; training	All
Added: stakeholder rules	Lead score 0–100, re-enquiry to previous agent, lead sharing, custom roles, Admin-only exports (section 10)	1

Section 12 of the brief also asks for recurring third-party costs (sections 1–3 and 10 of this document), direct versus provider-based integrations (section 3) and separate phase quotes (section 11).

16. Conclusion: final monthly cost

All figures in AED per month including 5% VAT. "Platform" is what we pay to run the CRM. "WhatsApp" is shown for information only: each company pays Meta directly, so our own cost is the platform line.

	LAUNCH	500 USERS	1 MILLION USERS
Who is using it	1 brokerage, 50 users, 5,000 leads	10 brokerages, 500 users, 50,000 leads	Mostly buyers and tenants, ~20,000 active a day, 400,000 leads
Platform (our cost)	**554**	**1,508**	**1,281**
WhatsApp (paid by companies to Meta)	**437**	**4,955**	**39,641**
Total incl. WhatsApp	**AED 991**	**AED 6,463**	**AED 40,922**
Platform cost per user	**AED 11.08**	**AED 3.02**	**under AED 0.01**

What changes at each size

• **500 users:** larger app servers and database, a small cache server, more email and error-tracking volume. Same design as launch; no rebuild.
• **1 million users:** the lean design in section 14: Cloudflare caching, image storage with no viewing fees, our own database servers with a live standby, self-hosted login and maps, and 3-year commitments.
• Cost per user falls sharply with scale because the core platform is shared by every brokerage and buyer.

Bottom line

Our own monthly platform cost is **AED 554** at launch, **AED 1,508** at 500 users and about **AED 1,281** at one million users, with data staying in the UAE. The 500-user figure still uses AWS's managed database; moving to the lean design at that point would bring it close to AED 1,000 as well. WhatsApp is paid by each company directly to Meta, so it never appears on our costs. Zoom, Calendly and respond.io are no longer needed. The one-million figure assumes mostly occasional users and should be re-checked before that scale.

Development is done in-house at **no cost**, over about 12 months, and version 1 launches with every feature. At AED 99 per user a month, running costs are covered by the first **6 paying users**, and at 500 users the CRM earns about AED 46,507 a month after running costs.